

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260519_232457_c
- started_at_local: 2026-05-19 23:24:57
- finished_at_local: 2026-05-19 23:57:54
- duration_hhmm: 00:33
- duration_seconds: 1976.961
- seed_offset: 2000
- replicate_label: c
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.08803	0.0	0.056019	0.737297	0.76	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235667	0.054968	0.121542	0.903863	0.44	False	0.004908	0.172081
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235667	0.054968	0.121542	0.901066	0.44	False	0.006066	0.244254
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.054968	0.121317	0.820961	0.4425	False	0.003143	-0.002078

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.888325	0.44	False	0.007131	0.718077
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.238386	0.064041	0.128273	0.893773	0.4425	False	0.001469	0.095546

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.08803, family holdout 0.0, combined 0.056019.
- Accepted novelty 0.737297, complexity 0.76, and adaptation efficiency 0.052001.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.903863, complexity 0.44, and adaptation efficiency 0.059402.
- Last-epoch runtime 60.9314 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.004908, positive scaffolds 1, Pareto runtime inflation 0.172081.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.149822, "gap_delta": 0.000224, "runtime_inflation": 0.172081, "same_gap_faster": false}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.901066, complexity 0.44, and adaptation efficiency 0.007624.
- Last-epoch runtime 160.58695 ms and distance evaluations 7185.75.

- Validation: surviving False, transplant delta 0.006066, positive scaffolds 1, Pareto runtime inflation 0.244254.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.182993, "gap_delta": 0.000224, "runtime_inflation": 0.244254, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.820961, complexity 0.4425, and adaptation efficiency 0.006693.
- Last-epoch runtime 74.8824 ms and distance evaluations 5058.75.
- Validation: surviving False, transplant delta 0.003143, positive scaffolds 0, Pareto runtime inflation -0.002078.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -0.002078, "same_gap_faster": true}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.888325, complexity 0.44, and adaptation efficiency 0.016341.
- Last-epoch runtime 75.207525 ms and distance evaluations 4823.25.
- Validation: surviving False, transplant delta 0.007131, positive scaffolds 1, Pareto runtime inflation 0.718077.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.866865, "gap_delta": -0.003304, "runtime_inflation": 0.718077, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.238386, family holdout 0.064041, combined 0.128273.
- Accepted novelty 0.893773, complexity 0.4425, and adaptation efficiency 0.005533.
- Last-epoch runtime 67.89985 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.001469, positive scaffolds 1, Pareto runtime inflation 0.095546.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 13, "complexity_score": 0.46, "distance_eval_inflation": 0.003718, "gap_delta": 0.006956, "runtime_inflation": 0.095546, "same_gap_faster": false}

Judge Appendix

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TSP Modular Operator Discovery Suite - Conservative Interpretation

Summary of Main Transfer Evidence (Held-out TSPLIB & Family Gap)

- Held-out TSPLIB gaps remain high (baseline ~0.235, operators ≥ 0.235 , except slight variations).

- Family holdout gaps (~0.05 - 0.06) show no clear improvement beyond baseline.
- Transfer gaps in modular operators range from approx. 0.118 to 0.128, not clearly better than baseline (baseline transfer gap ~0.121).

Operator Validation & Robustness

- No operator passes strict transplant or ablation checks convincingly:
- Ablation: Disabling the operator often does not hurt performance, indicating weak or no real effect.
- Transplant: Operators fail to transplant cleanly into critical scaffolds, e.g., `clustered_local_search`.
- Family tests: No operator demonstrates consistent benefit on held-out families; no positive family gains notable except one restart controller family, but with tradeoffs.

Detailed Notes on Key Operators

Candidate Pruner (e.g., "candidate_pruner_transfer_balanced", "descriptor_adaptive_pruner_for_cluster_grid_trap")

- Core idea: Adaptive pruning of candidate moves based on instance descriptors.
- Held-out TSPLIB gaps ~0.235 (no improvement over baseline).
- Transplant mean gap delta positive (worsened gaps) ~0.0049-0.006.
- Ablation shows disabled variant slightly better (negative gap delta vs full), so no ablation support for operator usefulness.
- Runtime inflation high (~17% to 24% on Pareto).
- Conclusion: Fails transplant and ablation criteria; no surviving candidate.

Perturbation Operator ("perturbation_double_bridge_escalating_2opt_escape_transfer")

- Core idea: Deterministic stagnation escape via structured perturbations.
- Family gap and held-out TSPLIB gap unchanged from baseline.
- Ablation variants equal to full operator; disabling does not hurt.
- Transplant shows no positive scaffold.
- Runtime slightly reduced but no benefit on transfer gap.
- Conclusion: No validation support; not a surviving candidate.

Restart Controller ("instance_adaptive_restart_controller_two_cluster_escape")

- Core idea: Adaptive restarts triggered by instance features to escape traps.
- Shows some family benefit on "two_cluster_bottleneck_tsp" (gap -0.024).
- However, held-out TSPLIB gap unchanged (0.235).
- Ablation suggests disabling operator worsens gap by +0.0033, some weak ablation support.
- Runtime inflation is high (~0.72).
- Transplant mean gap delta +0.007, runtime increase notable.
- Conclusion: Limited family transfer gain but fails held-out TSPLIB improvement and transplant; fails surviving candidate criteria.

Candidate Ranker ("rank_2opt_candidates_cluster_span_balanced")

- Core idea: Prioritizes local moves by score incorporating crossing and trap awareness.
- Family gap worsens (+0.064 gap), with severe regression on grid-like TSP.
- Ablation disabling operator reduces gap by -0.007, indicating operator hurts performance.
- Poor transplant results; no family-level improvement.
- Runtime inflation modest (~9.5%).
- Conclusion: Underperforms consistently; not surviving candidate.

Full-Solver Evolution (baseline reference)

- Best transfer and TSPLIB gaps: 0.056 (transfer) and 0.088 (TSPLIB) in 'phase7_full_solver_evolution', but no modular operator candidate survives.
- No modular operator achieved better gaps than full-solver baseline.

Pareto Tradeoff Assessment

- No operator yields Pareto improvements:
- None achieves better gap at equal or lower runtime.
- Most modular operators inflate runtime notably with no gap improvements.

Overall Conclusion

- ****No modular operator survives the strict transplant, ablation, family holdout, and held-out TSPLIB gap criteria to be considered interesting or reusable.****
- Weak or no operator effect indicated by ablation (disabling often does not harm or slightly improves).
- No operator demonstrates consistent improvements on held-out instance families, nor on TSPLIB benchmark.
- Some operators show minor family-level benefits at the cost of high runtime inflation and fail to transfer across scaffolds.
- Therefore, ****no claims about reusable modular operator structure or algorithm discovery are justified.****
- The full-solver evolution baseline remains the best performing condition but is not modularized.

Recommendations

- Focus further exploration on operators that:
- Show consistent positive gap reductions on held-out TSPLIB and family holdouts.
- Pass ablation checks with disabling clearly reducing performance.
- Successfully transplant into multiple scaffolds without severe runtime or gap regressions.
- Investigate runtime inflation drivers to improve Pareto efficiency.
- Avoid claims of algorithm discovery until modular operator validation passes scaffold and family tests.